CN Question Bank – Solutions

UNIT-1

Question 1 (2 Marks): Define Data Communication.

Answer:
Data communication is the process of transferring data between two or more devices through a transmission medium, such as a cable, fiber optic, or air. The communication can be between computers, mobile devices, or any electronic devices that need to exchange information. Essential components of data communication include the message, sender, receiver, transmission medium, and protocol.

Question 2 (2 Marks): Enlist the Applications of Computer Networks.

Answer:
Computer networks are widely used in various fields, such as:

1. **Resource Sharing**: Sharing of resources like printers, files, and applications among multiple users.
2. **Communication**: Email, video conferencing, and instant messaging for efficient communication.
3. **E-commerce**: Online shopping and transactions through secure networks.
4. **Data Transfer**: Efficient transfer of large amounts of data.
5. **Distributed Processing**: Allowing processes to run on different devices to enhance speed and efficiency.

Question 3 (2 Marks): Identify the Components of Data Communication.

Answer:
The main components of data communication are:

1. **Message**: The data or information to be communicated.
2. **Sender**: The device or entity that sends the data.
3. **Receiver**: The device or entity that receives the data.
4. **Transmission Medium**: The physical or wireless medium that carries the message.
5. **Protocol**: The set of rules that governs the data communication

Question 5 (2 Marks): Restate the Characteristics of Data Communication.

Answer:
The main characteristics of data communication include:

1. **Delivery**: Data should be delivered to the correct destination.
2. **Accuracy**: Data should be accurately transmitted without alteration.
3. **Timeliness**: Data should be delivered in a timely manner.
4. **Jitter**: Variation in packet arrival time should be minimal for smooth communication.

Question 6 (2 Marks): Define HUB in Computer Network.

Answer:
A **HUB** is a networking device that connects multiple devices in a network and allows data to be transmitted to all devices. It operates at the physical layer of the OSI

model and broadcasts the received signal to all connected devices, which may lead to network traffic issues.

Question 7 (2 Marks): State Characteristics of Computer Network.

Answer:
Characteristics of a computer network include:

1. **Scalability**: Networks can expand by adding more devices.
2. **Reliability**: Networks have mechanisms to ensure reliable communication.
3. **Security**: Networks provide data protection through encryption, firewalls, etc.
4. **Performance**: Speed and quality of data transfer are key performance factors.

Question 8 (2 Marks): Identify Attenuation in the Signal.

Answer:
Attenuation is the gradual loss of signal strength as it travels through a transmission medium. It is caused by factors such as distance and the medium's resistance, leading to signal degradation that requires amplification to maintain data integrity.

Question 9 (2 Marks): Define Topology with Example.

Answer:
Topology refers to the layout or arrangement of devices in a network. Examples include:

1. **Bus Topology**: All devices share a single communication line.
2. **Star Topology**: Each device is connected to a central hub.
3. **Mesh Topology**: Devices are interconnected, offering redundancy.

Question 10 (2 Marks): State the Data Representation Methods in Computer Network.

Answer:
Common data representation methods include:

1. **Text**: Using ASCII or Unicode to represent characters.
2. **Numbers**: Binary representation for digital transmission.
3. **Images**: Represented using pixels and stored as bitmaps.
4. **Audio and Video**: Converted into binary data using sampling.

Question 11 (2 Marks): Identify the Types of Network.

Answer:
The main types of networks are:

1. **PAN (Personal Area Network)**: Covers a small area, like a single room.
2. **LAN (Local Area Network)**: Covers a building or campus.
3. **MAN (Metropolitan Area Network)**: Covers a city.
4. **WAN (Wide Area Network)**: Covers large geographical areas, like countries.

Question 12 (2 Marks): Define Computer Network with its Application.

Answer:
A **Computer Network** is a group of interconnected devices that share resources and information. Applications include:

1. **File Sharing**: Accessing shared files over a network.
2. **Communication**: Email, video calls, and instant messaging.
3. **Resource Sharing**: Using networked printers and servers.

Question 13 (2 Marks): Enlist Characteristics of Data Communication.

Answer:
Characteristics of data communication include:

1. **Delivery**
2. **Accuracy**
3. **Timeliness**
4. **Jitter**
5. **Security**: Ensures data protection against unauthorized access.

Question 14 (2 Marks): State the Data Representation Technique.

Answer:
Data representation techniques include:

1. **ASCII**: For text representation.
2. **Binary**: For numbers.
3. **Pixels**: For images.
4. **Sampling**: For audio and video.

Question 15 (2 Marks): Calculate the Bandwidth of a Signal with Frequencies 256 Hz, 2000 Hz, 7000 Hz, and 356 Hz.

Solution:

1. **Bandwidth** = Highest Frequency - Lowest Frequency Bandwidth=7000 Hz−256 Hz=6744 HzBandwidth = 7000 \, \text{Hz} - 256 \, \text{Hz} = 6744 \, \text{Hz}Bandwidth=7000Hz−256Hz=6744Hz

Answer: Bandwidth = 6744 Hz

Question 16 (2 Marks): Calculate the Bandwidth of a Signal with Frequencies 9000 MHz, 4000 MHz, 2000 MHz, and 8000 MHz.

Solution:

1. **Bandwidth** = Highest Frequency - Lowest Frequency Bandwidth=9000 MHz−2000 MHz=7000 MHzBandwidth = 9000 \, \text{MHz} - 2000 \, \text{MHz} = 7000 \, \text{MHz}Bandwidth=9000MHz−2000MHz=7000MHz

Answer: Bandwidth = 7000 MHz

Question 17 (2 Marks): Calculate the Bandwidth of a Signal with Frequencies 3125 MHz, 254 MHz, 9000 MHz, and 200 MHz.

Solution: Bandwidth = Highest Frequency - Lowest Frequency
Bandwidth=9000 MHz−200 MHz=8800 MHzBandwidth = 9000 \, \text{MHz} - 200 \, \text{MHz} = 8800 \, \text{MHz}Bandwidth=9000MHz−200MHz=8800MHz

Answer: Bandwidth = 8800 MHz

Question 18 (4 Marks): Illustrate the Need for Computer Networks.

Answer:
Computer networks are essential for several reasons, including:

1. **Resource Sharing**: Networks allow sharing of hardware (like printers) and software applications, reducing costs.
2. **Data Sharing and Communication**: Users can exchange information instantly through email, chat, and video calls.
3. **Centralized Data Management**: Data can be stored on centralized servers, improving data management and security.
4. **Improved Efficiency and Productivity**: By enabling faster communication and collaboration, networks increase productivity.
5. **Remote Access**: Networks enable users to access resources and data remotely, facilitating flexibility and remote work.

Question 19 (4 Marks): Describe the Components of Data Communication.

Answer:
The essential components of data communication are:

1. **Message**: The actual data or information to be transmitted, which could be text, image, video, etc.
2. **Sender**: The device that initiates the transmission, such as a computer, phone, or server.
3. **Receiver**: The device that receives the message, interpreting and displaying it for the user.
4. **Transmission Medium**: The physical path through which the message travels, such as cables, fiber optics, or wireless signals.
5. **Protocol**: A set of rules and conventions that define the data transmission process to ensure accurate and secure communication.

Question 20 (4 Marks): Rewrite Data & Signal with its Types.

Answer:

- **Data**: Data represents information that needs to be communicated. It can exist in various forms:
  - **Analog Data**: Continuous data, like audio and video signals.
  - **Digital Data**: Discrete data, typically in binary form, representing text and images.
- **Signal**: A signal is the means of transmitting data over a medium. Signals can be of two main types:
  - **Analog Signal**: A continuous waveform used for audio and video transmission.
  - **Digital Signal**: A discrete waveform, used in computers and digital devices for binary data transmission.

Question 21 (4 Marks): Explain Data Flow with its Types.

Answer:
Data flow in computer networks refers to the direction in which data is transmitted between devices. The types of data flow include:

1. **Simplex**: Data flows in only one direction, from sender to receiver (e.g., TV broadcast).
2. **Half-Duplex**: Data can flow in both directions, but not simultaneously (e.g., walkie-talkies).
3. **Full-Duplex**: Data flows in both directions at the same time (e.g., phone calls, internet communication).

Question 22 (4 Marks): Discuss the Types of Computer Network Architecture.

Answer:
Computer network architecture refers to the design framework for network connectivity. The main types include:

1. **Peer-to-Peer (P2P)**: In this model, each device can act as both a client and a server. This architecture is suitable for small networks without centralized control.
2. **Client-Server**: A centralized server provides resources and services to multiple clients. It's commonly used in larger networks for better resource management and security.
3. **Hybrid**: Combines elements of both P2P and Client-Server architectures, allowing for flexible networking.

Question 23 (4 Marks): Write Network Hardware Components in Detail.

Answer:
Key hardware components of a network include:

1. **Router**: Directs data packets between different networks, connecting LANs and WANs.
2. **Switch**: Operates within a network, connecting devices and managing data traffic.
3. **Hub**: Basic device that connects multiple devices in a network, broadcasting data to all.
4. **Network Interface Card (NIC)**: A hardware component in devices that enables them to connect to a network.
5. **Modem**: Converts digital data to analog for transmission over phone lines and vice versa.
6. **Access Point**: Allows wireless devices to connect to a wired network.

Question 24 (7 Marks): Explain Topology with Its Different Types in Detail.

Answer:
Network topology refers to the layout of interconnected devices in a network. Types include:

1. **Bus Topology**: All devices are connected to a single central cable. It's simple but can have performance issues if many devices are active.
2. **Star Topology**: Devices connect to a central hub. It's reliable because failure in one cable doesn't affect others, but the hub is a single point of failure.

3. **Ring Topology**: Devices are connected in a circular format. Each device has exactly two neighbors. Data travels in one direction, making it easier to locate faults.
4. **Mesh Topology**: Each device is connected to every other device. It provides redundancy but is expensive due to the high number of connections.
5. **Tree Topology**: A combination of star and bus topologies. Devices are organized in a hierarchy, making it suitable for larger networks.
6. **Hybrid Topology**: A mix of two or more topologies to meet specific needs, often found in complex networks.

Question 25 (7 Marks): Illustrate the Types of Network in Detail.

Answer:

1. **Personal Area Network (PAN)**: Covers a small area, typically within a room, using devices like Bluetooth. Suitable for personal use.
2. **Local Area Network (LAN)**: Covers a single building or campus, connecting multiple computers. Commonly used in offices and homes.
3. **Metropolitan Area Network (MAN)**: Extends over a city, connecting multiple LANs. Used by organizations and ISPs to cover larger regions.
4. **Wide Area Network (WAN)**: Spans countries or continents, connecting multiple LANs and MANs. The internet is the largest WAN.
5. **Storage Area Network (SAN)**: A dedicated high-speed network providing access to consolidated storage resources.
6. **Enterprise Private Network (EPN)**: A private network managed by a business to connect various sites securely.

Question 26 (7 Marks): Discuss Computer Networks with Its Advantages and Disadvantages.

Answer:

- **Advantages**:
    1. **Resource Sharing**: Efficiently shares resources like printers and files, reducing costs.
    2. **Data Security and Management**: Centralized data storage improves data management and security.
    3. **Reliability**: Redundancy in networks ensures reliability and minimizes downtime.
    4. **Scalability**: Networks can easily expand by adding new devices.
    5. **Collaboration**: Networks facilitate collaboration through instant messaging and file sharing.
- **Disadvantages**:
    1. **Security Risks**: Vulnerable to unauthorized access and cyber threats.
    2. **Maintenance**: Requires constant maintenance and updates to stay secure.
    3. **Complex Setup**: Large networks are complex to set up and configure.

4. **Cost**: Hardware and software for a network can be expensive.
5. **Dependency on the Central Server**: In client-server models, server failure can disrupt the entire network.

Question 27 (7 Marks): Illustrate Network Hardware Components with Different Types of Connecting Devices.

Answer:
Network hardware components and connecting devices include:

1. **Routers**: Directs traffic between networks, connecting LANs and WANs. Types include wired and wireless routers.
2. **Switches**: Operates within a LAN to connect multiple devices. Types include managed and unmanaged switches.
3. **Hubs**: Simple devices that broadcast data to all connected devices. Variants include active and passive hubs.
4. **Bridges**: Connects two LANs to form a larger network, filtering traffic.
5. **Access Points**: Enables wireless devices to connect to a wired network, essential for Wi-Fi.
6. **Gateways**: Connects networks using different protocols, enabling data to flow between them.

Question 28 (7 Marks): Explain Topology with Its Different Types in Detail.

Answer:
Topology refers to the physical or logical arrangement of network devices. Types of network topologies include:

1. **Bus Topology**: All devices share a common backbone cable. It is cost-effective but has limited scalability and can be slow due to data collisions.
2. **Star Topology**: Each device is connected to a central hub. This setup is easy to troubleshoot, but the hub can be a single point of failure.
3. **Ring Topology**: Devices are connected in a circular format, where each device has two neighbors. Data travels in one direction, reducing collision risks, but one device failure can affect the entire network.
4. **Mesh Topology**: Every device is connected to all other devices. This provides redundancy and fault tolerance, but it is costly and complex to set up.
5. **Tree Topology**: Combines star and bus topologies in a hierarchical structure, useful for larger networks.
6. **Hybrid Topology**: Integrates multiple topologies to suit complex networking needs.

Question 29 (7 Marks): Discuss the Types of Computer Networks.

Answer:
The main types of computer networks are:

1. **Personal Area Network (PAN)**: Covers a small area, typically within an individual's reach, like Bluetooth connections for personal devices.
2. **Local Area Network (LAN)**: Spans a building or campus, used in schools, homes, or offices. It is

commonly wired but can also be wireless.
3. **Metropolitan Area Network (MAN)**: Covers a larger area, such as a city, connecting multiple LANs. It is often used by ISPs and municipalities.
4. **Wide Area Network (WAN)**: Covers extensive geographical areas, connecting multiple LANs and MANs. The internet is the largest WAN.
5. **Enterprise Private Network (EPN)**: A private network for a business organization, connecting its various offices securely.

Question 30 (7 Marks): Illustrate Data Communication with Its Characteristics and Components in Detail.

Answer:
Data Communication is the exchange of data between devices over a medium, ensuring the data reaches the correct destination in a timely, accurate manner. Characteristics include:

1. **Delivery**: Data should be delivered to the correct destination.
2. **Accuracy**: Data must be transmitted without errors.
3. **Timeliness**: Data must arrive promptly for real-time applications.
4. **Jitter**: Variation in packet arrival times should be minimized.

Components of data communication include:

1. **Message**: The information to be transmitted.
2. **Sender**: The device that initiates the communication.
3. **Receiver**: The destination device for the data.
4. **Transmission Medium**: The physical or wireless path for data transmission.
5. **Protocol**: Rules governing the data transmission process.

Question 31 (12 Marks): Explain the Following Types of Network: PAN, LAN, MAN, WAN.

Answer:

1. **Personal Area Network (PAN)**: Covers a small area around a person (e.g., Bluetooth devices). Used for connecting personal devices and has limited range and speed.
2. **Local Area Network (LAN)**: Spans a building or campus, commonly found in schools and businesses. Allows high-speed data transfer and resource sharing within a limited area.
3. **Metropolitan Area Network (MAN)**: Covers a city or metropolitan area. Used by ISPs to provide city-wide internet and connect LANs. MANs typically use high-speed cables like fiber optics.
4. **Wide Area Network (WAN)**: Spans a large geographical area, connecting multiple LANs and MANs, often through leased telecom lines or satellites. WANs, like the internet, support long-distance data transfer.

Question 32 (12 Marks): Describe Computer Network with Its Types of Architecture, Advantages, and Disadvantages in Detail.

Answer:
A **Computer Network** is an interconnected system of devices that share resources and communicate. Types of network architecture:

1. **Peer-to-Peer (P2P)**: Each device can act as both a client and server. It is simple, low-cost, and suited for small networks, but lacks centralized control and can be less secure.
2. **Client-Server**: A centralized server provides resources to clients. It is scalable, has centralized security, and is used in larger networks but can have higher costs and is dependent on server uptime.

Advantages:

1. **Resource Sharing**: Centralized access to resources like printers and storage.
2. **Cost-Efficiency**: Reduces the need for multiple devices by sharing resources.
3. **Remote Access**: Enables access to data and resources from different locations.
4. **Improved Communication**: Facilitates instant messaging, emails, and video calls.

Disadvantages:

1. **Security Risks**: Vulnerable to hacking and malware.
2. **Maintenance**: Requires regular updates and management.
3. **Complexity**: Large networks are complex to set up and manage.
4. **Server Dependency**: In client-server architectures, server failure can disrupt the network.

Q33: Differentiate between the following

a) Data vs. Signal

Data refers to the raw information or content that is intended for communication, such as text, audio, video, or images. It represents the actual message that needs to be transmitted from a sender to a receiver.

A signal, on the other hand, is the physical form of data as it travels over a medium. Signals are used to encode and transmit the data across network channels. Signals can be in analog form, like continuous electromagnetic waves, or digital form, like discrete voltage pulses. Essentially, while data is the information, the signal is the means of transferring that information over a network.

Example: Text data is transmitted as binary digital signals over a network.

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b) PAN vs. LAN vs. MAN vs. WAN

A **Personal Area Network (PAN)** covers a very small area, typically within a few meters, and is mainly used for connecting personal devices such as smartphones, tablets, and laptops. Examples include Bluetooth and USB connections. PANs are useful for connecting devices without a larger network and are often used for short-distance communication.

A **Local Area Network (LAN)** covers a limited area, such as a single building or campus. It connects devices like computers, printers, and servers within an office or home. LANs are typically fast and use technologies like Ethernet or Wi-Fi to facilitate high-speed data exchange among devices within close proximity.

A **Metropolitan Area Network (MAN)** spans a larger area than a LAN, typically covering a city or large campus. MANs connect multiple LANs and can be used by organizations that need network access across various sites within a metropolitan region. Examples of MANs include citywide Wi-Fi networks and cable television networks.

A **Wide Area Network (WAN)** covers a very large geographic area, often extending across countries or continents. WANs connect multiple LANs and MANs over long distances, allowing for interconnection across cities, states, or countries. The Internet is the largest example of a WAN, facilitating global communication.

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c) Connectionless vs. Connection-Oriented Service

In a **connectionless service**, data is sent without establishing a dedicated path or session between the sender and receiver. Each data packet is sent independently, with no guarantee of delivery or order. This type of service is faster and has less overhead since it doesn't involve setting up or maintaining a connection. **UDP (User Datagram Protocol)** is a common protocol used in connectionless services, making it suitable for applications that prioritize speed, like voice-over-IP (VoIP) and online gaming, where delays are critical.

In contrast, a **connection-oriented service** requires establishing a connection before data transfer begins. This ensures a reliable path between sender and receiver, providing higher reliability as it guarantees the delivery and correct sequence of data packets. **TCP (Transmission Control Protocol)** is a widely used connection-oriented protocol, ideal for applications needing reliable communication, such as file transfers and email, where data accuracy and completeness are essential.

Connection-oriented services have more overhead than connectionless services because of the additional steps involved in connection setup, data transfer, and disconnection.

UNIT-2

Question 34 (2 Marks): Find the Class of the Following IP Address:

- **a) 11000001 10000011 00011011 11111111**
- **b) 252.5.15.111**

Solution:

1. **Class Identification**:
    - **a) 11000001 10000011 00011011 11111111**:
        - The first bits are `110`, so this is a **Class C** IP address.
    - **b) 252.5.15.111**:
        - The first octet, `252`, falls in the range `240–255`, indicating a **Class E** IP address (reserved for experimental purposes).

Question 35 (2 Marks): List Out the Transmission Media in Computer Network.

Answer:
Transmission media in computer networks can be categorized as:

1. **Wired (Guided) Media**:
   - **Twisted Pair Cable**: Commonly used in LANs.
   - **Coaxial Cable**: Used in cable television and older Ethernet networks.
   - **Fiber Optic Cable**: High-speed, long-distance transmission.
2. **Wireless (Unguided) Media**:
   - **Radio Waves**: Used for Wi-Fi and cellular networks.
   - **Microwaves**: Used in satellite communication.
   - **Infrared**: Short-range communication, such as remote controls.

Question 36 (2 Marks): Restate Propagation Delay.

Answer:
Propagation Delay is the time taken for a signal to travel from the sender to the receiver over a transmission medium. It depends on the distance between the two devices and the speed of the signal through the medium. Propagation delay can be calculated by:

Propagation Delay=DistancePropagation Speed\text{Propagation Delay} = \frac{\text{Distance}}{\text{Propagation Speed}}Propagation Delay=Propagation SpeedDistance

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Question 37 (2 Marks): Give an Example of Simple Parity Check.

Answer:
In a **Simple Parity Check**, an additional bit, called the parity bit, is added to data to make the total number of `1`s even (even parity) or odd (odd parity).

Example:

- Data: `1010`
- For even parity, add a `0` to make the total number of `1`s even: `10100`
- For odd parity, add a `1` to make the total number of `1`s odd: `10101`

Question 38 (2 Marks): Write a Short Note on Transmission Impairment.

Answer:
Transmission Impairment refers to the degradation of signal quality during transmission. Types include:

1. **Attenuation**: Loss of signal strength over distance.
2. **Distortion**: Alteration of the signal's original shape due to different frequencies traveling at different speeds.
3. **Noise**: Unwanted interference that distorts the signal, such as thermal noise or electromagnetic interference.

Question 39 (2 Marks): Define Data Rate Limit.

Answer:
The **Data Rate Limit** is the maximum rate at which data can be transmitted over a communication channel. Factors influencing data rate include bandwidth, signal-to-noise

ratio, and error rates. Shannon's and Nyquist's formulas are often used to determine the data rate limits in noisy and noiseless channels, respectively.

Question 40 (2 Marks): Enlist the Types of Multiple Access Protocol.

Answer:
Multiple Access Protocols allow multiple devices to share the same communication medium. Types include:

1. **Time Division Multiple Access (TDMA)**: Divides the channel into time slots.
2. **Frequency Division Multiple Access (FDMA)**: Divides the channel into different frequency bands.
3. **Code Division Multiple Access (CDMA)**: Assigns unique codes to each user for simultaneous data transmission.
4. **Carrier Sense Multiple Access (CSMA)**: Devices sense the medium before transmitting.

Question 41 (2 Marks): State the Services of Data Link Layer.

Answer:
The **Data Link Layer** provides services such as:

1. **Framing**: Organizing data into frames for transmission.
2. **Error Detection and Correction**: Identifying and correcting errors that occur during transmission.
3. **Flow Control**: Ensuring data flow is at a manageable rate for the receiver.
4. **Medium Access Control (MAC)**: Determining how devices access the shared communication medium.

Question 42 (2 Marks): Find the Class of IP Address 252.5.10.23.

Solution:
The first octet is `252`, which falls within the range of **Class E** IP addresses (240-255). Class E is reserved for experimental purposes.

Question 43 (2 Marks): Restate the Definition of IP Address with Example.

Answer:
An **IP Address** is a unique identifier assigned to each device on a network, used to locate and identify the device. IP addresses are usually expressed in dotted decimal notation for IPv4 (e.g., `192.168.1.1`). They can belong to various classes (A, B, C, D, and E) based on network requirements.

Question 44 (4 Marks): Calculate the Number of Signal Levels Required to Send 165 Kbps Over a Noiseless Channel with a Bandwidth of 20 KHz.

Solution: Using the **Nyquist Theorem** for a noiseless channel:

Data Rate=2×Bandwidth×log2L\text{Data Rate} = 2 \times \text{Bandwidth} \times \log_2 LData Rate=2×Bandwidth×log2L

Where:

- Data Rate = 165 Kbps
- Bandwidth = 20 KHz

Rearrange to solve for LLL:

L=2Data Rate2×Bandwidth=2165,0002×20,000L = 2^{\frac{\text{Data Rate}}{2 \times \text{Bandwidth}}} = 2^{\frac{165,000}{2 \times 20,000}}L=22×BandwidthData Rate =22×20,000165,000 L=24.125≈17L = 2^{4.125} \approx 17L=24.125≈17

Answer: Approximately 17 signal levels are required.

Question 45 (4 Marks): Calculate the Number of Signal Levels Required to Send 265 Kbps Over a Noiseless Channel with a Bandwidth of 20 KHz.

Solution: Using the **Nyquist Theorem**:

L=2Data Rate2×Bandwidth=2265,0002×20,000L = 2^{\frac{\text{Data Rate}}{2 \times \text{Bandwidth}}} = 2^{\frac{265,000}{2 \times 20,000}}L=22×BandwidthData Rate =22×20,000265,000 L=26.625≈100L = 2^{6.625} \approx 100L=26.625≈100

Answer: Approximately 100 signal levels are required.

Question 46 (4 Marks): Consider a Noisy Channel with a Bandwidth of 3000 Hz Transmitting a Signal with Two Signal Levels. What Can Be the Maximum Bitrate?

Solution: Using **Shannon's Theorem** for a noisy channel:

Capacity=Bandwidth×log2(1+SNR)\text{Capacity} = \text{Bandwidth} \times \log_2 (1 + \text{SNR})Capacity=Bandwidth×log2(1+SNR)

With 2 signal levels (binary), SNR can be calculated if known, but commonly:

Capacity=3000×log2(1+SNR)\text{Capacity} = 3000 \times \log_2 (1 + \text{SNR})Capacity=3000×log2(1+SNR)

Question 47 (4 Marks): Calculate Whether 192.1.1.47 to 192.1.1.60 is a Valid IP Address Block or Not.

Solution:

1. **Determine if the Range Lies Within a Valid Subnet**:
   - The IP range 192.1.1.47 to 192.1.1.60 falls under Class C (192.0.0.0 to 223.255.255.255).
   - Class C addresses typically have a subnet mask of 255.255.255.0, meaning each subnet can have 256 addresses (including network and broadcast addresses).
2. **Check the Continuity of IP Addresses**:
   - The given range (47 to 60) contains 14 addresses, which is valid within a Class C subnet.

Answer: Yes, 192.1.1.47 to 192.1.1.60 is a valid IP address block within a Class C subnet.

Question 48 (4 Marks): If We Need to Send 365 Kbps Over a Noiseless Channel with a Bandwidth of 30 KHz, How Many Signal Levels Do We Need?

Solution: Using the **Nyquist Theorem**:

L=2Data Rate2×Bandwidth=2365,0002×30,000L = 2^{\frac{\text{Data Rate}}{2 \times \text{Bandwidth}}} = 2^{\frac{365,000}{2 \times 30,000}}L=22×BandwidthData Rate =22×30,000365,000 L=26.083≈68L = 2^{6.083} \approx 68L=26.083≈68

Answer: Approximately 68 signal levels are required.

Question 49 (4 Marks): Find the Corresponding Binary Pattern for the Following CRC Generators:

- **a) x7+x6+x4+x3+x+1x^7 + x^6 + x^4 + x^3 + x + 1x7+x6+x4+x3+x+1**
- **b) x7+x6+x3+x2+xx^7 + x^6 + x^3 + x^2 + xx7+x6+x3+x2+x**
- **c) x5+x4+x2+1x^5 + x^4 + x^2 + 1x5+x4+x2+1**
- **d) x10+x7+x4+x3+x+1x^{10} + x^7 + x^4 + x^3 + x + 1x10+x7+x4+x3+x+1**

Solution: Convert each polynomial to binary representation, where the power of xxx represents the position with `1`.

1. **a)** x7+x6+x4+x3+x+1x^7 + x^6 + x^4 + x^3 + x + 1x7+x6+x4+x3+x+1 = `11011011`
2. **b)** x7+x6+x3+x2+xx^7 + x^6 + x^3 + x^2 + xx7+x6+x3+x2+x = `11001110`
3. **c)** x5+x4+x2+1x^5 + x^4 + x^2 + 1x5+x4+x2+1 = `110101`
4. **d)** x10+x7+x4+x3+x+1x^{10} + x^7 + x^4 + x^3 + x + 1x10+x7+x4+x3+x+1 = `10001011011`

Question 50 (4 Marks): Restate the Switching Techniques with Its Types.

Answer:
Switching techniques determine how data is routed from sender to receiver. The main types are:

1. **Circuit Switching**: Establishes a dedicated path for the entire communication session (e.g., traditional telephone networks).
2. **Packet Switching**: Divides data into packets, which are routed independently across the network (e.g., the internet).
3. **Message Switching**: The entire message is stored and forwarded, requiring intermediate storage at each switch.

Question 51 (4 Marks): Check Whether 10.2.20.32 to 10.2.20.39 is a Valid IP Address Block or Not.

Solution:

1. **Class and Subnet Verification**:
   - The IP range 10.2.20.32 to 10.2.20.39 falls under Class A (1.0.0.0 to 126.0.0.0).
   - Class A default subnet mask is 255.0.0.0, allowing for larger networks.
2. **IP Range Check**:
   - The range from 32 to 39 includes eight addresses, valid for a subnet division.

Answer: Yes, 10.2.20.32 to 10.2.20.39 is a valid IP address block.

Question 52 (4 Marks): Define Transmission Impairment with Its Types.

Answer:
Transmission Impairment is the degradation of signal quality during transmission. Types include:

1. **Attenuation**: Loss of signal strength due to distance.
2. **Distortion**: Changes in signal shape as different frequencies travel at different speeds.
3. **Noise**: Unwanted signals that interfere with data transmission, including thermal, impulse, and crosstalk noise.

Question 53 (4 Marks): Write a Short Note on Framing Method.

Answer:
Framing is the process of dividing data into manageable units or frames for transmission. Methods include:

1. **Character Count**: Frames include a count of characters, which defines their length.
2. **Flag Bytes with Byte Stuffing**: Uses special bytes to mark frame boundaries, adding escape characters when data matches flag patterns.
3. **Starting and Ending Flags with Bit Stuffing**: Flags are represented by unique bit patterns, with extra bits added to data to avoid misinterpretation.
4. **Physical Layer Coding Violations**: Uses specific signals or voltage changes to denote the start/end of frames.

Question 54 (4 Marks): Discuss the Transmission Mode with Its Types in Computer Network.

Answer:
Transmission Modes determine the direction and timing of data flow:

1. **Simplex**: Data flows in only one direction, from sender to receiver (e.g., keyboard to computer).
2. **Half-Duplex**: Data flows in both directions, but not simultaneously (e.g., walkie-talkies).
3. **Full-Duplex**: Data flows in both directions simultaneously (e.g., telephone communication).

Question 55 (4 Marks): Illustrate the Transmission Medium with Its Types in Detail.

Answer:
Transmission media are physical pathways to transmit data:

1. **Wired Media**:
   - **Twisted Pair Cable**: Common in LANs; includes shielded (STP) and unshielded (UTP).
   - **Coaxial Cable**: Used in cable TV and older Ethernet.
   - **Fiber Optic Cable**: High-speed transmission over long distances, using light signals.
2. **Wireless Media**:
   - **Radio Waves**: Used in Wi-Fi, with frequencies below 1 GHz.
   - **Microwaves**: Used for long-distance, line-of-sight communication.
   - **Infrared**: Used in short-range applications, like remote controls.

Question 56 (4 Marks): Discuss Error Detection Methods in Computer Network.

Answer:
Error detection methods identify errors during data transmission:

1. **Parity Check**: Adds a parity bit to make the number of 1s even or odd.
2. **Checksum**: Adds all data bytes, with the sum transmitted to verify data integrity.
3. **Cyclic Redundancy Check (CRC)**: Uses polynomial division to generate a CRC code, which is checked at the receiver.
4. **Hamming Code**: Adds redundancy bits to detect and correct single-bit errors.

Question 57 (4 Marks): Find the Classes of Given IP Address.

- **a) 251.10.2.27**
- **b) 20.10.1.10**
- **c) 11111101 00011101 10101010 11100101 10101010**
- **d) 10111000 11100011 00010101 10110001 11001010**

Solution:

1. **a) 251.10.2.27**: First octet `251` indicates **Class E** (240-255).
2. **b) 20.10.1.10**: First octet `20` indicates **Class A** (1-126).
3. **c) 11111101**: Binary prefix `1111` indicates **Class E**.
4. **d) 10111000**: Binary prefix `101` indicates **Class B**.

Question 58 (4 Marks): Sketch Switching Techniques in Computer Network.

Answer:
Switching techniques are methods to route data through a network:

1. **Circuit Switching**: Establishes a dedicated path; all data follows this path (e.g., telephone networks).
2. **Packet Switching**: Data is split into packets; each packet takes the best route independently (e.g., internet).
3. **Message Switching**: Entire message is sent to each intermediate node and stored temporarily, then forwarded.

Question 59 (7 Marks): Discuss the Switching Technique with Its Types in Detail.

Answer:
Switching techniques enable efficient data routing:

1. **Circuit Switching**:
   - A dedicated path is established for the entire communication.
   - **Advantages**: Consistent connection quality.
   - **Disadvantages**: Inefficient for short messages, as the path remains reserved.
2. **Packet Switching**:
   - Data is divided into packets, which are sent independently.
   - **Advantages**: Efficient and flexible; suitable for large networks.
   - **Disadvantages**: Packets can arrive out of order, requiring reassembly.
3. **Message Switching**:
   - The whole message is stored temporarily at each switch before forwarding.
   - **Advantages**: No need for a dedicated path.
   - **Disadvantages**: Increased delay due to storage requirements.

Question 60 (7 Marks): Explain the Types of Error.

Answer:
Errors in data transmission can be categorized as:

1. **Single-Bit Error**: Only one bit is altered.
2. **Burst Error**: Multiple bits in a sequence are altered.
3. **Random Error**: Errors occur sporadically.
4. **Systematic Error**: Errors occur in a pattern due to system flaws.

Question 61 (7 Marks): Examine the IP Address in Computer Network.

Answer:
An **IP Address** is a unique numerical label assigned to each device connected to a network that uses the Internet Protocol for communication. It serves two main functions:

1. **Identification of Host or Network Interface**: An IP address uniquely identifies a device on a network.
2. **Location Addressing**: It provides the location of the device in the network.

IP addresses are of two main types:

1. **IPv4**: A 32-bit address divided into four octets, typically written in dotted decimal format (e.g., 192.168.1.1). It includes five classes (A, B, C, D, E), each designated for different uses.
2. **IPv6**: A 128-bit address written in hexadecimal format, designed to overcome IPv4 limitations by providing a larger address space.

Each IP address consists of two parts:

- **Network Part**: Identifies the specific network.
- **Host Part**: Identifies the device within the network.

Question 62 (7 Marks): Calculate the Message Transmitted by the Sender Using CRC Generator and Check if the Receiver Receives It Correctly.

- Message: `100100`
- Generator Polynomial: x3+x2+1x^3 + x^2 + 1x3+x2+1

Solution:

1. **Convert Generator Polynomial**: x3+x2+1x^3 + x^2 + 1x3+x2+1 becomes `1101`.
2. **Append Zeros**: Add three zeros to the message to make `100100000`.
3. **Divide Message by Generator**:
   - Perform binary division and find the remainder.
4. **Transmit Message**: Append the remainder to the original message for transmission.
5. **Check at Receiver**: The receiver performs division using the same generator. If the remainder is zero, the message was received correctly.

Question 63 (7 Marks): Explain IP Address with Types of Classful Internet Address in Detail.

Answer:
An **IP Address** is a unique identifier for a device on a network. It helps route data to the correct destination. In **Classful Addressing**, IP addresses are divided into classes:

1. **Class A**:
   - Range: `1.0.0.0` to `126.0.0.0`

- For large networks with many hosts.
- Default Subnet Mask: `255.0.0.0`

2. **Class B**:
   - Range: `128.0.0.0` to `191.255.0.0`
   - For medium-sized networks.
   - Default Subnet Mask: `255.255.0.0`
3. **Class C**:
   - Range: `192.0.0.0` to `223.255.255.0`
   - For small networks with fewer hosts.
   - Default Subnet Mask: `255.255.255.0`
4. **Class D**:
   - Range: `224.0.0.0` to `239.255.255.255`
   - Reserved for multicasting.
5. **Class E**:
   - Range: `240.0.0.0` to `255.255.255.255`
   - Reserved for experimental purposes.

Question 64 (7 Marks): Illustrate the Error Detection & Correction Methods in Detail.

Answer:
Error detection and correction methods ensure data integrity during transmission.

1. **Parity Check**:
   - Adds a parity bit to make the count of `1`s even (even parity) or odd (odd parity).
   - Detects single-bit errors but does not correct them.
2. **Checksum**:
   - Sums all data segments and includes the sum in the transmission.
   - The receiver verifies by summing and comparing.
3. **Cyclic Redundancy Check (CRC)**:
   - Uses polynomial division to generate a remainder, which is transmitted with the data.
   - The receiver uses the same polynomial to detect errors.
4. **Hamming Code**:
   - Adds redundancy bits to data, allowing detection and correction of single-bit errors.
   - Works by positioning redundancy bits in powers of two.
5. **Forward Error Correction (FEC)**:
   - Allows the receiver to detect and correct errors without retransmission.
   - Common in real-time systems, using techniques like Reed-Solomon coding.

Question 65 (12 Marks): Calculate Whether the Following IP Address Blocks Are Valid:

- **a) 100.1.2.32 to 100.1.2.42**
- **b) 192.1.1.47 to 192.1.1.60**
- **c) 25.1.3.22 to 125.1.3.86**

Solution:

1. **For 100.1.2.32 to 100.1.2.42**:
   - **Class**: Class A (100.0.0.0 to 126.0.0.0)
   - **Range Validity**: The range is valid within Class A.
2. **For 192.1.1.47 to 192.1.1.60**:
   - **Class**: Class C (192.0.0.0 to 223.255.255.0)
   - **Range Validity**: The range is valid within Class C.
3. **For 25.1.3.22 to 125.1.3.86**:
   - **Class**: Ranges across Class A and B.

- **Range Validity**: Invalid, as IP ranges cannot span multiple classes.

UNIT-3

66. State the layers of the TCP/IP Model. (2 marks)

The TCP/IP model has four layers:

1. **Application Layer**: It provides services for network applications, such as HTTP, FTP, SMTP.
2. **Transport Layer**: It ensures end-to-end communication, with protocols like TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).
3. **Internet Layer**: Responsible for logical addressing and routing with protocols like IP (Internet Protocol).
4. **Network Interface Layer**: It handles the physical transmission of data over network hardware, defining protocols for the hardware connection.

---

67. Find the shortest path of the given graph from Source 's' to Destination 'e'. (2 marks)

For solving shortest path problems, algorithms like **Dijkstra's Algorithm** are used to find the minimum cost path between two nodes in a graph. If the graph is provided, you can follow this method to calculate it.

---

68. Rewrite Shortest Path Routing. (2 marks)

Shortest Path Routing refers to routing methods that determine the shortest possible path between a source and a destination node in a network. It is used to optimize data flow by minimizing delay, cost, or distance. Algorithms like **Dijkstra's Algorithm** and **Bellman-Ford Algorithm** are typically employed to find the shortest path.

---

69. Define Flooding in a computer network. (2 marks)

Flooding is a simple routing technique in which every incoming packet is sent out on every outgoing line except the one it arrived on. This ensures that the packet reaches all parts of the network. Although it guarantees delivery, it is not efficient because it can lead to network congestion due to duplicate packets.

---

70. Define Network Layer. (2 marks)

The **Network Layer** is the third layer of the OSI model and is responsible for packet forwarding, including routing through intermediate routers. It manages logical addressing (IP addresses) and is crucial for transferring data across networks. Protocols in this layer include **IP**, **ICMP**, and **IGMP**.

---

71. Enlist the functions of the Network Layer. (2 marks)

Key functions of the network layer include:

1. **Routing**: Determines the best path for data packets to travel.
2. **Logical Addressing**: Assigns IP addresses to devices for identification and communication.
3. **Packet Forwarding**: Transfers packets from one network to another.
4. **Fragmentation and Reassembly**: Breaks down larger packets into smaller ones for transmission and reassembles them at the destination.

---

72. State the types of Congestion Control Algorithms. (2 marks)

Congestion control algorithms help manage traffic in a network to prevent congestion. Types include:

1. **Leaky Bucket Algorithm**: Controls the rate at which data packets are sent into the network.
2. **Token Bucket Algorithm**: Allows bursty data to be sent at higher rates while controlling the average rate of data flow.
3. **Congestion Avoidance Algorithms**: Such as **TCP Tahoe**, **TCP Reno**, **AIMD (Additive Increase Multiplicative Decrease)**.

---

73. Identify the protocols of the Network Layer. (2 marks)

Common protocols used in the network layer include:

1. **IP (Internet Protocol)**: Provides logical addressing and routing of packets.
2. **ICMP (Internet Control Message Protocol)**: Used for error reporting and diagnostics.
3. **IGMP (Internet Group Management Protocol)**: Manages multicast group memberships.
4. **ARP (Address Resolution Protocol)**: Maps IP addresses to MAC addresses.

---

74. Find the shortest path of the given tree. (2 marks)

To find the shortest path in a tree structure, you would typically traverse the tree using methods like **Breadth-First Search (BFS)** or **Depth-First Search (DFS)**, depending on the provided graph. The shortest path is usually the one with the least weight or hops between nodes.

---

75. Illustrate the Network Layer Design Issues in detail. (4 marks)

Network layer design issues focus on the following:

1. **Routing**: Determining the optimal path for packet transfer across networks.
2. **Congestion Control**: Managing network traffic to avoid congestion and packet loss.
3. **Packet Forwarding**: Ensuring packets are correctly forwarded to their destination through routers.
4. **Addressing and Naming**: Assigning logical addresses (IP addresses) for devices to ensure proper identification and communication.

5. **Error Handling**: Detecting and correcting errors in packet transmission.
6. **Fragmentation and Reassembly**: Breaking down large packets for easier transmission and reassembling them at the destination.

---

76. Write a short note on Flooding Routing Algorithm. (4 marks)

Flooding is a routing algorithm in which every incoming packet is sent out on every outgoing link, except the one it came from. While this method ensures that a packet reaches its destination, it has several drawbacks:

- **No routing tables** are required, which simplifies the process.
- However, it leads to **excessive duplication** of packets, resulting in network congestion.
- It can also cause **infinite looping**, although techniques like **Time-to-Live (TTL)** can be used to prevent this.

77. Write a short note on Congestion. (4 marks)

Congestion occurs in a network when the demand for resources (such as bandwidth) exceeds the available capacity, leading to a slowdown in network performance. Causes of congestion include high data traffic, slow links, and inefficient routing.
Consequences of congestion:

- **Packet loss**: Due to buffer overflow at routers.
- **Increased delay**: Packets have to wait longer in queues, resulting in delayed transmissions.
- **Jitter**: Variation in packet delay, affecting real-time services like video calls.

Methods to control congestion:

1. **Congestion Avoidance Algorithms**: Such as TCP's congestion control mechanisms like slow start and congestion window.
2. **Queue Management**: Techniques like Random Early Detection (RED) help prevent congestion by dropping packets early.
3. **Load Balancing**: Distributes traffic across multiple paths to avoid overloading any single route.

---

78. Find Distance Vector Routing of following routers weights. (7 marks)

Distance Vector Routing is a dynamic routing protocol in which each router calculates the best route to each destination based on information from its neighbors. Routers exchange distance vectors periodically to keep routing tables updated. The **Bellman-Ford algorithm** is often used in this process.

To find the solution, you would:

1. Initialize each router with its direct distance to neighbors.
2. Update the routing table by calculating new paths based on the information received from adjacent routers.
3. Continue updating until all routers have converged on the shortest paths.

If an example graph with routers and weights is provided, I can apply this process step-by-step.

---

79. Describe the Routing Algorithms with its types in detail. (7 marks)

Routing algorithms are used to find the optimal path for data packets to travel across a network. The main types include:

1. **Static Routing**:
    - Routes are manually configured and do not change unless manually updated.
    - Suitable for small networks where routes rarely change.
2. **Dynamic Routing**:
    - Routes are automatically adjusted based on network conditions.
    - Examples include **RIP (Routing Information Protocol)**, **OSPF (Open Shortest Path First)**, and **BGP (Border Gateway Protocol)**.
    - Dynamic routing protocols can be further classified into:
        - **Distance Vector Routing**: Routers share information with neighbors (e.g., RIP).
        - **Link-State Routing**: Each router has complete information about the network topology (e.g., OSPF).
3. **Flooding**: Sends packets through all possible paths. It's used when a network has no knowledge of routes, though it leads to high redundancy and congestion.
4. **Hierarchical Routing**: Divides the network into layers to manage large networks efficiently, reducing the complexity of routing tables.

---

80. Consider the following network; three routers are there: A, B, and C. Find the routing table for all the routers A, B, and C using Distance Vector Routing. (7 marks)

To find the routing table using **Distance Vector Routing**:

1. **Initialization**: Each router starts with a table containing only the distances to its immediate neighbors. Non-neighbor distances are set to infinity.
2. **Exchange**: Each router periodically sends its distance vector to its neighbors.
3. **Update**: Upon receiving distance vectors from neighbors, each router updates its own table by comparing the new information to its current routing table.
4. **Iteration**: The process repeats until no further updates are needed.

Example: Let’s say A, B, and C are connected in a triangle with certain link costs. The routing table for each router will eventually stabilize, showing the shortest paths to all other routers.

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81. Describe Routing Algorithms with its types in detail. (7 marks)

This is a repeat of **Question 79**, where routing algorithms are described in detail.

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82. Restate the Services provided by Network Layer. (7 marks)

The network layer provides the following services:

1. **Logical Addressing**: Assigns logical addresses (IP addresses) to each device in a network for identification.
2. **Routing**: Determines the best path for data to travel from source to destination.
3. **Packet Forwarding**: Transfers data packets from one network to another, making use of routing tables.
4. **Error Handling**: Detects and sometimes corrects errors that may occur in data transmission.
5. **Fragmentation and Reassembly**: Large packets are broken down into smaller packets for transmission and are reassembled at the destination.
6. **Internetworking**: Connects different types of networks, enabling data to travel across diverse networks.

---

83. Distinguish between Connection-less vs Connection-oriented Networks. (7 marks)

Feature	Connection-less Network	Connection-oriented Network
Setup	No setup is required before sending data (e.g., UDP).	A connection must be established before data can be sent (e.g., TCP).
Reliability	Less reliable; packets may arrive out of order or get lost.	More reliable; guarantees data delivery in order.
Overhead	Lower overhead, as no connection maintenance is needed.	Higher overhead due to connection setup and tear-down.
Use Case	Suitable for real-time applications like video streaming.	Suitable for applications that require reliable data transfer like email or file transfer.
Examples	UDP (User Datagram Protocol)	TCP (Transmission Control Protocol)

---

84. Determine the Shortest Path of given graph from Source A to Destination F. (7 marks)

The **Dijkstra's Algorithm** or **Bellman-Ford Algorithm** can be used to determine the shortest path in a graph. The basic steps involve:

1. Initializing distances from the source node (A) to all other nodes as infinity, except for the source itself.
2. Visiting each neighboring node and updating the shortest known distance.
3. Repeating the process until the shortest path to the destination node (F) is determined.

If a specific graph is provided, the exact path can be calculated by tracing through the steps of these algorithms.

---

85. Describe OSI Reference Model & TCP/IP Reference Model with its layers in details. (12 marks)

OSI (Open Systems Interconnection) Model:

1. **Physical Layer**: Transmits raw bits over a physical medium.
2. **Data Link Layer**: Responsible for node-to-node data transfer and error correction.
3. **Network Layer**: Handles routing and logical addressing (IP addresses).
4. **Transport Layer**: Ensures reliable data transfer (e.g., TCP, UDP).
5. **Session Layer**: Manages sessions between applications.
6. **Presentation Layer**: Translates data formats (e.g., encryption, compression).
7. **Application Layer**: Provides services for applications like email, file transfer, etc.

TCP/IP Model:

1. **Network Interface Layer**: Combines OSI's Physical and Data Link layers.
2. **Internet Layer**: Corresponds to the OSI's Network Layer, handles IP addressing and routing.
3. **Transport Layer**: Same as the OSI's Transport Layer, responsible for reliable communication.
4. **Application Layer**: Combines OSI's Session, Presentation, and Application layers, and provides services for end-user applications.

Key differences:

- OSI has 7 layers, while TCP/IP has 4.
- OSI is a theoretical model, while TCP/IP is a practical framework used in real networks.

---

86. Describe the classification of Switching Techniques. (12 marks)

Switching techniques allow data to be transmitted over a network by efficiently utilizing network resources. The three main types of switching techniques are:

1. **Circuit Switching**:
    - A dedicated communication path is established between two devices before data can be transferred.
    - The entire bandwidth of the communication link is reserved for the connection.
    - Used in traditional telephone networks.
    - Example: Public Switched Telephone Network (PSTN).
    - **Advantages**: Guaranteed continuous data transmission.
    - **Disadvantages**: Inefficient for data that doesn't need a constant connection.
2. **Packet Switching**:
    - Data is divided into smaller packets, and each packet is transmitted independently over the network.

- Packets may take different paths to the destination and are reassembled upon arrival.
- Used in the internet and most data communication networks.
- Example: Internet Protocol (IP).
- **Advantages**: Efficient use of network resources and better for bursty traffic.
- **Disadvantages**: Packets can arrive out of order or get lost.

3. **Message Switching**:
   - Entire messages are transferred from one node to another, stored at intermediate points, and forwarded when a path is available.
   - There is no need for a dedicated path.
   - **Advantages**: Does not require establishing a connection before transmitting data.
   - **Disadvantages**: Longer delays due to the need for complete message storage at each node.

---

87. Explain OSI Reference Model in a computer network. (12 marks)

The **OSI (Open Systems Interconnection)** model is a conceptual framework used to understand and implement standards for network communication. It divides the networking process into seven distinct layers, each with its own function:

1. **Physical Layer**:
   - Deals with the physical connection between devices, including the transmission of raw bitstreams over a medium.
   - Components: Cables, switches, and physical network interfaces.
2. **Data Link Layer**:
   - Responsible for reliable data transfer between two physically connected devices.
   - Handles error detection, frame synchronization, and flow control.
   - Example protocols: Ethernet, PPP (Point-to-Point Protocol).
3. **Network Layer**:
   - Manages the routing and forwarding of data packets between devices across different networks.
   - Provides logical addressing (IP addresses).
   - Example protocols: IP, ICMP.
4. **Transport Layer**:
   - Ensures end-to-end data transmission, reliability, and flow control.
   - Example protocols: TCP (Transmission Control Protocol), UDP (User Datagram Protocol).
5. **Session Layer**:
   - Manages sessions or connections between applications.
   - Establishes, maintains, and terminates communication sessions.
6. **Presentation Layer**:
   - Translates data between the application layer and the network format.
   - Handles data encryption, compression, and format

conversion (e.g., ASCII, JPEG).

7. **Application Layer**:
    - Provides network services directly to user applications.
    - Example protocols: HTTP, FTP, SMTP.

UNIT-4

88. Classify the Elements of Transport Protocols. (2 marks)

Transport protocols are responsible for delivering data across networks reliably. The key elements include:

1. **Port Numbers**: Identifies the source and destination applications.
2. **Connection Establishment**: Handles the setup of a connection between devices (e.g., TCP's three-way handshake).
3. **Flow Control**: Ensures data is sent at a rate that the receiver can handle (e.g., sliding window protocol).
4. **Error Control**: Ensures error detection and correction to provide reliable communication.
5. **Congestion Control**: Manages traffic load to prevent network congestion.

---

89. Explain TCP Header with its Message Format. (4 marks)

The **TCP header** is part of each TCP segment and contains critical information for managing a connection:

1. **Source Port (16 bits)**: Identifies the application sending data.
2. **Destination Port (16 bits)**: Identifies the application receiving data.
3. **Sequence Number (32 bits)**: The order number of the first byte in this segment, crucial for reassembling data in the correct sequence.
4. **Acknowledgment Number (32 bits)**: The next expected byte from the sender, used to acknowledge received segments.
5. **Data Offset (4 bits)**: Specifies the size of the TCP header.
6. **Reserved (3 bits)**: Reserved for future use, must be set to zero.
7. **Flags (9 bits)**: Controls the connection (e.g., SYN, ACK, FIN, URG).
8. **Window Size (16 bits)**: Controls flow, indicating the number of bytes the sender can transmit before waiting for acknowledgment.
9. **Checksum (16 bits)**: Ensures the integrity of the TCP header and data.
10. **Urgent Pointer (16 bits)**: Points to the urgent data in the segment (used when URG flag is set).

---

90. Discuss in detail TCP Timer with its types. (4 marks)

TCP uses several timers to manage the connection and ensure reliable data transfer:

1. **Retransmission Timer**:
    - Triggered when a segment is sent, waiting for an acknowledgment (ACK). If no ACK is received within the timeout, the segment is retransmitted.
2. **Persistence Timer**:
    - Prevents a deadlock when the receiver advertises a zero window size. The persistence timer probes the receiver until the window size increases.
3. **Keepalive Timer**:
    - Used to detect if the connection is still alive. If no activity is detected for a specified time, TCP sends keepalive probes. If no response is received, the connection is closed.
4. **Time-Wait Timer**:
    - After a connection is closed, this timer keeps the connection in a wait state to ensure all delayed packets are correctly handled and any duplicate segments are discarded.

---

91. List out the Elements of Transport Protocols. (4 marks)

The core elements of transport protocols include:

1. **Port Numbers**: Identifies specific processes or applications on the sender's and receiver's machines (e.g., HTTP uses port 80).
2. **Sequence Numbers**: Ensures ordered delivery by assigning a unique number to each segment of data.
3. **Acknowledgments (ACKs)**: Confirms the successful receipt of data segments.
4. **Flow Control**: Prevents the sender from overwhelming the receiver (e.g., using a sliding window protocol).
5. **Congestion Control**: Manages the data flow to avoid overwhelming the network with too much traffic.
6. **Error Detection**: Detects and corrects errors in the data transmission, ensuring reliability.

---

92. Write Sliding Window Protocol. (4 marks)

The **Sliding Window Protocol** is a flow control mechanism used to manage the transmission of data between sender and receiver. Key features include:

1. **Window Size**: Defines the number of unacknowledged packets that can be in transit before requiring acknowledgment from the receiver.
2. **Sender**: Transmits multiple packets without waiting for individual acknowledgments but ensures they stay within the window size.
3. **Receiver**: Receives packets and sends acknowledgments when required, allowing the sender to slide the window forward.
4. **Efficiency**: The protocol increases efficiency by allowing continuous data transmission instead of waiting for an acknowledgment after each packet.

---

93. State Functions of Presentation Layer. (4 marks)

The **Presentation Layer** is responsible for formatting data to ensure that it can be understood by both the application sending the data and the one receiving it. Key functions include:

1. **Data Translation**: Converts data between different formats (e.g., EBCDIC to ASCII).
2. **Data Encryption/Decryption**: Secures the data during transmission and decrypts it at the destination.
3. **Data Compression**: Reduces the size of the data to optimize bandwidth usage during transmission.
4. **Data Serialization**: Converts complex data structures into a format that can be easily transmitted and reassembled.

---

94. Rewrite Types of TCP Timers. (4 marks)

The main TCP timers include:

1. **Retransmission Timer**: Ensures reliable data transmission by retransmitting unacknowledged packets after a timeout period.
2. **Persistence Timer**: Prevents deadlocks when the receiver's window size is zero, probing the receiver periodically.
3. **Keepalive Timer**: Detects whether the connection is still alive by sending keepalive messages if no data has been exchanged for a set time.
4. **Time-Wait Timer**: Ensures that duplicate segments do not interfere with a new connection by keeping the connection in a wait state after closing.

---

95. Distinguish between Go-Back-N ARQ and Selective Repeat ARQ. (4 marks)

Feature	**Go-Back-N ARQ**	**Selective Repeat ARQ**
Error Recovery	Retransmits all packets from the error point.	Retransmits only the erroneous packet.
Buffering	Receiver does not buffer out-of-order packets.	Receiver buffers out-of-order packets.
Efficiency	Less efficient due to unnecessary retransmissions.	More efficient as only erroneous packets are retransmitted.
Complexity	Simpler to implement.	More complex due to the need for out-of-order handling.

Go-Back-N ARQ and **Selective Repeat ARQ** are protocols that help ensure reliable data transmission by dealing with errors.

1. **Go-Back-N ARQ**:
   In Go-Back-N, the sender can send several frames before needing an acknowledgment, up to a set "window size." If an error occurs in a frame, the receiver discards that frame and all frames that follow. The sender then "goes back" to the frame with the error and resends it along

with all subsequent frames. This method is simple but can waste time by resending frames that may have been received correctly.

2. **Selective Repeat ARQ**:
   In Selective Repeat, the sender also sends multiple frames within a window. However, if a frame has an error, only that specific frame is resent. The receiver can store frames that arrive out of order until the missing frame is corrected. This makes Selective Repeat more efficient, especially when errors are frequent, as it reduces unnecessary retransmissions.

Main Differences:

- **Go-Back-N** resends all frames after an error, while **Selective Repeat** only resends the frame with the error.
- Go-Back-N is simpler but can be less efficient; Selective Repeat requires more memory but is more efficient in handling errors.

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96. Discuss the Retransmission with its types in detail. (7 marks)

Retransmission is a mechanism in transport protocols used to resend lost or corrupted data packets to ensure reliability. Types of retransmission include:

1. **Go-Back-N Retransmission**:
   - If an error is detected, the sender retransmits all packets starting from the erroneous one, even if subsequent packets were received correctly.
   - Simple but inefficient for high-error networks.
2. **Selective Repeat Retransmission**:
   - Only the packets identified as lost or erroneous are retransmitted, making it more efficient than Go-Back-N.
   - The receiver must have a buffer to store out-of-order packets.
3. **Fast Retransmission**:
   - Instead of waiting for the timeout, the sender retransmits the missing packet as soon as it receives three duplicate ACKs, indicating packet loss.

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97. Write in detail the 3-way and 4-way handshake in TCP with a diagram. (7 marks)

3-Way Handshake (Connection Establishment):

1. **SYN**: The client sends a SYN (synchronize) packet to the server to initiate the connection.
2. **SYN-ACK**: The server responds with a SYN-ACK (synchronize-acknowledgment) packet to acknowledge the client's SYN request and to synchronize with the client.
3. **ACK**: The client sends an ACK packet back to the server, confirming the establishment of the connection.

4-Way Handshake (Connection Termination):

1. **FIN**: The client sends a FIN (finish) packet to terminate the connection.

2. **ACK**: The server acknowledges the FIN request with an ACK.
3. **FIN**: The server sends a FIN packet to the client to terminate its end of the connection.
4. **ACK**: The client acknowledges the server's FIN request, completing the termination.

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98. Analyze TCP Header Format in Transport Layer Protocol. (7 marks)

The **TCP header** is crucial for managing data transmission in the transport layer. Important fields in the TCP header include:

1. **Source Port**: Identifies the application on the sending host.
2. **Destination Port**: Identifies the application on the receiving host.
3. **Sequence Number**: Keeps track of the order of segments.
4. **Acknowledgment Number**: Confirms receipt of data.
5. **Data Offset**: Indicates where the TCP header ends and the data begins.
6. **Control Flags**: SYN, ACK, FIN, URG, PSH, RST flags control the state of the connection.
7. **Window Size**: Manages flow control by indicating how much data the receiver can accept.
8. **Checksum**: Ensures data integrity.
9. **Urgent Pointer**: Points to urgent data when the URG flag is set.

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99. Explain with diagram Connection Establishment and Termination in TCP. (7 marks)

TCP (Transmission Control Protocol) Connection Establishment and Termination

TCP connection management includes two primary processes: connection establishment and connection termination.

1. Connection Establishment (Three-Way Handshake)

The **three-way handshake** process is used to establish a reliable TCP connection between a client and a server. This process includes the following steps:

- **Step 1: SYN (Synchronize)**
  The client initiates the connection by sending a **SYN** (synchronize) packet to the server. This packet indicates that the client intends to start a communication session and includes a sequence number.
- **Step 2: SYN-ACK (Synchronize-Acknowledge)**
  The server responds to the client's SYN packet with a **SYN-ACK** packet. This packet acknowledges the client’s request by sending back an acknowledgment number (client's sequence number + 1) and includes its own sequence number.
- **Step 3: ACK (Acknowledge)**
  The client replies with an **ACK** packet, acknowledging the server's SYN-ACK. This final acknowledgment completes the connection setup, and the client and server are now ready to start data transfer.

2. Connection Termination (Four-Way Handshake)

The **four-way handshake** process is used to terminate an established TCP connection.

Each side of the connection (client and server) terminates independently, allowing bidirectional communication to close gracefully.

- **Step 1: FIN (Finish)**
  The client initiates termination by sending a **FIN** packet to the server, indicating that it has finished sending data.
- **Step 2: ACK (Acknowledge)**
  The server acknowledges the client's FIN by sending an **ACK** packet. However, the server can still send data until it's ready to close.
- **Step 3: FIN (Finish)**
  When the server is ready to terminate the connection, it sends a **FIN** packet back to the client.
- **Step 4: ACK (Acknowledge)**
  The client acknowledges the server's FIN with an **ACK** packet, completing the connection termination process.

Summary

- **Three-Way Handshake**:
  Establishes a connection, with SYN, SYN-ACK, and ACK packets.
- **Four-Way Handshake**: Terminates the connection, involving FIN and ACK packets from both client and server.

This structured handshake process ensures reliable communication by setting up and gracefully closing the TCP connection, ensuring that all data is transmitted accurately before termination.

UNIT-5

100. List out the Application Layer Services. (2 Marks)

Answer:
The Application Layer in networking provides various services that allow applications to communicate across networks. These services include:

1. **File Transfer** (e.g., FTP)
2. **Email Services** (e.g., SMTP)
3. **Remote Access** (e.g., Telnet)
4. **Web Browsing** (e.g., HTTP)
5. **Directory Services** (e.g., DNS)
6. **Network Management** (e.g., SNMP)

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101. Write the Responsibilities of Presentation Layer. (2 Marks)

Answer:
The Presentation Layer is responsible for:

1. **Data Translation**: Converts data from application format to network format and vice versa.
2. **Data Encryption/Decryption**: Ensures secure data transmission by encoding and decoding information.
3. **Data Compression/Decompression**: Reduces the amount of data to be transmitted, improving efficiency.

4. **Data Formatting**: Manages formats such as ASCII, JPEG, etc., for compatibility between different systems.

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102. Enlist Functions of Application Layer. (2 Marks)

Answer:
The Application Layer functions include:

1. **Network Virtual Terminal**: Allows remote login capabilities.
2. **File Transfer, Access, and Management (FTAM)**: Provides services for file manipulation on remote systems.
3. **Mail Services**: Facilitates email forwarding and storage.
4. **Directory Services**: Manages distributed database sources and resource lookup.

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103. Classify the different Security Services in network. (2 Marks)

Answer:
Network security services can be classified as:

1. **Confidentiality**: Protects data from unauthorized access.
2. **Integrity**: Ensures data has not been altered.
3. **Authentication**: Verifies the identity of the communicating entities.
4. **Non-repudiation**: Ensures that the sender cannot deny sending a message.
5. **Access Control**: Limits access to network resources.

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104. Write short note on Digital Signature. (2 Marks)

Answer:
A **Digital Signature** is a cryptographic mechanism that verifies the authenticity and integrity of a digital message or document. Created using the sender's private key, it allows the receiver to confirm the message's origin and content using the sender's public key. Digital signatures are widely used in electronic transactions, ensuring non-repudiation.

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105. Examine the Architecture of WWW. (4 Marks)

Answer:
The World Wide Web (WWW) architecture consists of three primary components:

1. **Uniform Resource Locator (URL)**: A unique address for resources on the web.
2. **Hypertext Transfer Protocol (HTTP)**: The protocol used to transfer web pages.
3. **Web Browser and Web Server**: The client-server model where a web browser (client) requests information from the web server which processes and serves the request.

The WWW operates on a client-server architecture where:

- **Clients** (e.g., browsers) request resources,

- **Servers** store, retrieve, and send the requested resources to clients,
- Resources are typically HTML files, images, or multimedia content, often linked through hypertext.

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106. Write short note on WWW and HTTP. (4 Marks)

Answer:

- **WWW (World Wide Web)**: A global collection of interconnected documents and multimedia content accessible via the Internet. It operates on hyperlinks and is accessed using web browsers.
- **HTTP (Hypertext Transfer Protocol)**: The foundation of data communication on the WWW. HTTP defines how messages are formatted and transmitted, and how web servers and browsers should respond to requests. It operates as a request-response protocol between client and server, typically using port 80 for HTTP and port 443 for HTTPS (secure version).

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107. Describe HTTP with different headers. (7 Marks)

Answer:
HTTP, or Hypertext Transfer Protocol, enables communication between clients and servers. It includes the following headers:

1. **General Headers**: Provide common information applicable to both request and response (e.g., Date, Connection).
2. **Request Headers**: Information sent by the client to the server (e.g., Accept, Host, User-Agent).
3. **Response Headers**: Details from the server to the client about the response (e.g., Server, Set-Cookie).
4. **Entity Headers**: Describe the body of the request or response (e.g., Content-Length, Content-Type).

Examples of HTTP Headers:

- **GET**: Requests data from a specified resource.
- **POST**: Submits data to be processed to a specified resource.
- **PUT**: Updates a current resource with new data.
- **DELETE**: Deletes the specified resource.

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108. Explain Symmetric and Asymmetric Key Cryptography in detail. (7 Marks)

Answer:

1. **Symmetric Key Cryptography**:
    - **Description**: Uses a single, shared key for both encryption and decryption.
    - **Advantages**: Faster and efficient for large data transfers.
    - **Disadvantages**: Requires secure key distribution, which can be challenging.
    - **Examples**: DES, AES.
2. **Asymmetric Key Cryptography**:

- **Description**: Utilizes a pair of keys—a public key for encryption and a private key for decryption.
- **Advantages**: Provides better security without needing to share the private key.
- **Disadvantages**: Slower than symmetric encryption.
- **Examples**: RSA, ECC.

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109. Distinguish between the following: (12 Marks)

- **a) Symmetric Key vs. Asymmetric Key**
- **b) Connectionless vs. Connection-Oriented Networks**
- **c) TCP vs. OSI**

Answer:

- **a) Symmetric Key vs. Asymmetric Key**:
  - **Symmetric Key**: Single key for encryption and decryption; faster; key distribution is a challenge.
  - **Asymmetric Key**: Pair of public and private keys; secure key exchange; slower but more secure.
- **b) Connectionless vs. Connection-Oriented Networks**:
  - **Connectionless**: Data sent without establishing a connection (e.g., UDP); faster, but less reliable.
  - **Connection-Oriented**: Requires a connection before data transfer (e.g., TCP); reliable, error-checked data.
- **c) TCP vs. OSI**:
  - **TCP/IP Model**: Has four layers (Application, Transport, Internet, Network Access); simpler and widely used.
  - **OSI Model**: Seven layers (Application to Physical); theoretical, providing a detailed standard reference.

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110. Rewrite about following: (12 Marks)

- **a) SMTP**
- **b) TFTP**
- **c) DNS**
- **d) FTP**

Answer:

- **a) SMTP (Simple Mail Transfer Protocol)**: Used to send and forward emails between servers and clients over the Internet.
- **b) TFTP (Trivial File Transfer Protocol)**: Simplified file transfer protocol with no authentication, used for simple, small file transfers.
- **c) DNS (Domain Name System)**: Translates human-readable domain names (e.g., www.example.com) into IP addresses, allowing easier access to resources.
- **d) FTP (File Transfer Protocol)**: Allows file exchange between a

client and a server over a network with login and authentication.

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111. Explain the following layer (any 4): (12 Marks)

- **a) Physical Layer**
- **b) Data Link Layer**
- **c) Network Layer**
- **d) Transport Layer**
- **e) Session Layer**
- **f) Presentation Layer**

Answer (Choose any four):

- **Physical Layer**: Responsible for the transmission of raw data bits over a physical medium, including electrical and mechanical specifications.
- **Data Link Layer**: Provides error detection and correction, organizes data into frames, and manages node-to-node communication.
- **Network Layer**: Handles routing and forwarding, defines logical addressing (e.g., IP addresses), and manages data transfer between networks.
- **Transport Layer**: Ensures end-to-end data integrity, flow control, and error recovery, with protocols like TCP and UDP.
- **Session Layer**: Manages sessions or connections between applications, handling setup, maintenance, and termination.
- **Presentation Layer**: Transforms data into an application-readable format, handling data translation, encryption, and compression.